

$2^{-2} = \frac{1}{2^2}$
 $\rightarrow 1) n(-ve) \rightarrow +ve$
 $2) \text{reciprocal of } 2$
 $\hookrightarrow x = \frac{1}{x}$

$n = -2$
 $2^{-2} = \frac{1}{2^2}$
 $\rightarrow n \rightarrow -n$
 $x \rightarrow \frac{1}{x}$

$x^n \rightarrow x \times x \times x \dots \times x$
 ntimes
 $\hookrightarrow \text{pow} = 1;$
 $\hookrightarrow \text{pow} = \text{pow} \times x;$

$n \rightarrow -ve$
 $n = -n;$
 $x = \frac{1}{x};$
 $n = -2$
 $-(2)$
 $x = 2$
 $2^{-2} = \frac{1}{2^2}$

math. pow

```

if(n < 0) {
    n = -n;
    x = 1/x;
}
for(int i = 1; i <= n; i++) {
    pow = pow * x;
}
return pow;
    
```

$x = 2, n = 16$
 $2^{16} \rightarrow (2^2)^8$
 $(4)^8 \rightarrow (4^2)^4$
 $(16)^4 \rightarrow (16^2)^2$
 $(256)^2 \rightarrow (256^2)^1$

$(165536)^1 \rightarrow \text{return}$

$2^{15} \rightarrow (2^2)^7 \times 2$
 $(4)^7 \times 2$
 $(4^2)^3 \times 2$
 $(16)^3 \times 2$

$\text{pow} = \text{pow} \times n$

x^n
 $\rightarrow \text{even}$
 $\rightarrow \text{even}$

long m = n;

```

long m = n; // if n
if (m < 0) {
    m = -m;
    x = 1/x;
}

```

```

double pow = 1;
while (m > 0) {
    if ((m % 2 != 0)) pow = pow * x;
}

```

```

x = x * x;
m = m / 2;
}

```

```

return pow;

```

$(16)^3$

$16 \times (16)^2$

$(256)^2$

min $\rightarrow$ +ve

- 2147483648 @ - - - - - 2147483647

n =